MATH0037 Logic

Year:	2026–2027
Code:	MATH0037
Level:	6 (UG), 6 (PG)
Normal student group(s):	UG: Y3 Mathematics programmes PG: MSc Mathematics
Value:	15 credits (7.5 ECTS credits)
Term:	1
Assessment:	90% examination, 10% coursework
Normal Pre-requisites:	At least one of the following is recommended: MATH0028, MATH0029, MATH0034, MATH0051, MATH0053.
Lecturer:	Dr S Coskey

Course Description and Objectives

This module is an introduction to propositional and first order logic. We study the interplay between semantic (truth, meaning) and syntactic (deductive) aspects of logic. The module has an emphasis on applications of ideas from logic in other areas of mathematics, mainly graph theory, combinatorics, and algebra.

Recommended Texts

- HB Enderton, *A Mathematical Introduction to Logic* (Harcourt)
- D van Dalen, *Logic and Structure* (Springer)
- I Chiswell and W Hodges, *Mathematical Logic* (Oxford)
- GS Boolos and RC Jeffrey, *Computability and Logic* (Cambridge)

Detailed Syllabus

- **Propositional Logic**
  - The language of propositional logic.
  - Propositional functions, truth values, valuations, compactness and applications.
  - Rooted trees and Kőnig's lemma.
  - The strengthened Ramsey theorem.
  - Deductions (Hilbert or natural)
- **First order Logic**
  - Concrete examples of first order languages, theories, and their models, with particular emphasis on graphs, orders, Peano Arithmetic and other algebraic structures.
  - Consistency and Godel's completeness theorem.
  - Elementary equivalence.
  - The Łoś-Vaught test.
  - Nonstandard models of arithmetic.

- **Basic Set Theory and Countability**
  - Countable and uncountable sets.
  - The Cantor-Schroeder Bernstein theorem.
  - Cantor's diagonalization argument.
  - The theory of dense linear orders without endpoints.